

Corrections to *Gödel Without (Too Many) Tears*

These are corrections for the currently available print-on-demand paperback and hardback versions of *GWT*, and the currently downloadable PDF.

A major glitch

pp. 28 ff. On p. 28, the language of baby arithmetic, L_B is said to have just negation as its only propositional connective. But the very next page, in Schema 2, the conditional creeps in; on p. 32, the lower proof, there are wffs containing the conditional. But then on p. 33 it is said that the only wffs of BA are equations preceded by zero or more negation signs, and the conditional is forgotten about again. Ouch! [In seeking to simplify a proof in *IGT*, I carelessly introduced this incoherence.]

Option (a) is to replace Schema 2 with a corresponding rule of inference ‘from a wff of the form $S\zeta = S\xi$ infer $\zeta = \xi$ ’, and then make the obvious adjustments to the lower proof on p. 32. This would slightly spoil the way that BA and Q are set up in parallel, but otherwise keeps things simple. (We’d have to change some later wording in obvious ways to fit.)

Option (b) is to officially build the conditional into the language L_B (and we might as well add the other connectives), and make the logic of BA a full truth-functional propositional logic, plus the laws of identity. But then the proof of Theorem 11 on p. 33 will need to be revised to allow for the presence of connectives in addition to negation. This is in fact how things were done in earlier versions of *GWT*. So we can argue like this

Our Theorems 9 and 10 now enable us to prove the following:

Theorem. *Suppose σ and τ are terms of L_B . Then if $\sigma = \tau$ is true, then $BA \vdash \sigma = \tau$, and if $\sigma = \tau$ is false, then $BA \vdash \sigma \neq \tau$.*

Proof Let σ evaluate to s and τ evaluate to t . Then, by Theorem 9, (i) $BA \vdash \sigma = \bar{s}$ and (ii) $BA \vdash \tau = \bar{t}$.

Now, suppose $\sigma = \tau$ is true. Then $s = t$, and so $\bar{s}$ must be the very same numeral as $\bar{t}$. We can therefore immediately conclude from (i) and (ii) that $BA \vdash \sigma = \tau$ by the logic of identity.

Suppose, on the other hand, that $\sigma = \tau$ is false, so $s \neq t$. Then by Theorem 10, $BA \vdash \bar{s} \neq \bar{t}$, and together with (i) and (ii) that implies $BA \vdash \sigma \neq \tau$, again by the logic of identity. $\square$

And from that last theorem, it more or less immediately follows that

Theorem. *BA is negation complete.*

Proof The only atomic claims expressible in BA are equations involving terms; all other sentences are truth-functional combinations of such atoms. But we’ve just seen that BA (1) proves each true atom and (2) proves the negation of each false atom.

Now we can just appeal to a standard result about propositional logic. Suppose a wff φ is built up using the truth-functional connectives from certain atoms. Take a valuation of those atoms. Then if φ is true on that valuation, then φ can be proved by propositional logic from the true atoms and the negations of the false atoms. While if φ is false on that valuation, then $\neg\varphi$ can be proved from the true atoms and the negations of the false atoms.

Hence, since for *any* sentence φ of BA, the theory can prove its true atoms and the negations of its false atoms, the logic of BA can then be used to derive φ if it is true and $\neg\varphi$ otherwise. Hence BA is negation complete. $\square$

Possibly distracting typos

p. 43. There are some numbering foul-ups in the displayed proof. With changes from (what it is now) line 6, it should read

- | | | |
|-----|--|----------------------------------|
| 1. | $0 + 0 = 0$ | Instance of Q's Axiom 4 |
| 2. | $0 + a = a$ | Supposition |
| 3. | $S(0 + a) = Sa$ | From 2 by the identity laws |
| 4. | $0 + Sa = S(0 + a)$ | Instance of Q's Axiom 5 |
| 5. | $0 + Sa = Sa$ | From 3, 4 |
| 6. | $0 + a = a \rightarrow 0 + Sa = Sa$ | From 2–5 by Conditional Proof |
| 7. | $\forall x(0 + x = x \rightarrow 0 + Sx = Sx)$ | From 6, since a was arbitrary. |
| 8. | $0 + 0 = 0 \wedge \forall x(0 + x = x \rightarrow 0 + Sx = Sx)$ | From 1, 7 |
| 9. | $\{0 + 0 = 0 \wedge \forall x(0 + x = x \rightarrow 0 + Sx = Sx)\} \rightarrow \forall x(0 + x = x)$ | Instance of Induction Schema |
| 10. | $\forall x(0 + x = x)$ | From 8, 9 by Modus Ponens |

p. 71, line 5. k_{Is} should be k_{Si} .

p. 48, line 12. Delete the parenthetical ‘(for quantifier-free σ and τ)’. Because σ and τ stand in for terms which are by definition quantifier-free.

p. 107, lines 5, 4 and 3 up from bottom. Each occurrence of Prf on those lines should be overlined, $\overline{\text{Prf}}$. Also, on the last of these lines, $\overline{m}$ should be a quantified variable (and to better match Def, 54, use ‘ w ’). So:

Hence, because $\overline{\text{Prf}}$ captures $\overline{\text{Prf}}$, we have each of $T \vdash \neg \overline{\text{Prf}}(0, \overline{n})$, $T \vdash \neg \overline{\text{Prf}}(1, \overline{n})$, $T \vdash \neg \overline{\text{Prf}}(2, \overline{n})$, $\dots$, $T \vdash \neg \overline{\text{Prf}}(\overline{m}, \overline{n})$.

Therefore, we will also have (ii) $T \vdash (\forall w \leq \overline{m}) \neg \overline{\text{Prf}}(w, \overline{n})$ – since Q and hence T ‘knows’ about bounded quantifiers (see §7.1).

Trivial typos

Page	Line	Is	Should be
13	5	needs to expressively rich	needs to be expressively rich
50	8	that that	that
53	–7	Gödel first	Gödel’s first
58	–6	We can now says	We can now say
72	Defn 18 ii.	$\neg \varphi((\overline{m}, \overline{n}))$	$\neg \varphi(\overline{m}, \overline{n})$
82	14	$\varphi(\overline{\ulcorner n \urcorner})$	$\varphi(\overline{n})$
131	13	$\wedge \neg \text{Prf}_T(a, \overline{\ulcorner \perp \urcorner})$,	$\wedge \neg \overline{\text{Prf}}_T(a, \overline{\ulcorner \perp \urcorner})$,

With thanks to David Furcy, SangYoon Lee, Kyungnam Moon, David Lloyd Owen and Sherman Page. Particular thanks to Ben Selfridge for pointing out the major glitch.